Data Visualization

CSC 3115 - Advanced Programming

Why data visualization?

Objective: Simplify the understanding of raw data

Importance in web applications

❖ Engaging users

❖ Deriving insights

❖ Decision-making

❖ Story-telling

Examples: analytics dashboards, financial reports, real-time monitoring.

Basic principles

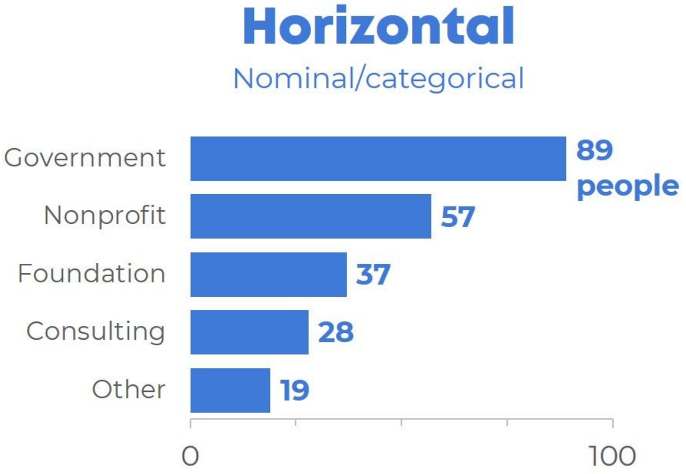
- ❖ **Clarity:** clear and easily understood.
- ❖ **Simplicity:** minimize clutter, use clear labels and legends, limit your colour palette, focus on one main message.
- ❖ **Accuracy:** should correctly reflect the underlying data.
- ❖ **Efficiency:** Convey insights as quickly as possible. Maximize the data-ink ratio by minimizing non-essential elements.
- ❖ **Consistency:** Maintain a uniform design style.

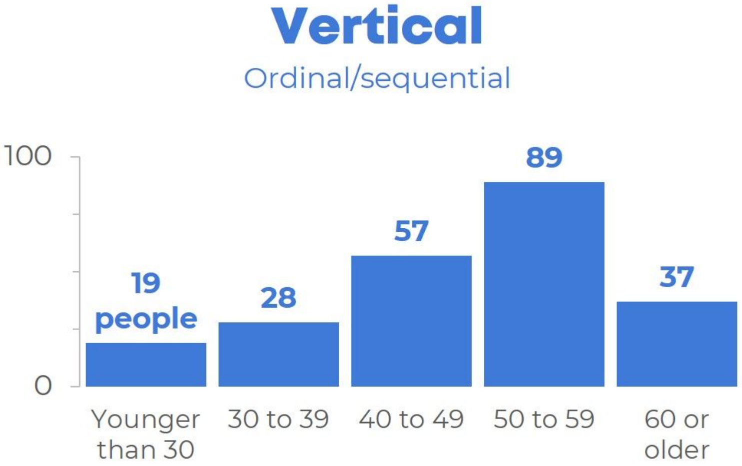
Choosing the right visualization

- ❖ Who is your target audience?
- ❖ What are you trying to communicate?

Goal	**Recommended chart**	**Example**
Compare values	Bar, Column	Sales by region
Show trends	Line, Area	Temperature over time
Composition	Pie, Doughnut	Market share
Distribution	Histogram, Box plot	Test scores
Relationship	Scatter, Bubble	Height vs Weight
Flow/Process	Sankey diagram	Clinical pathway

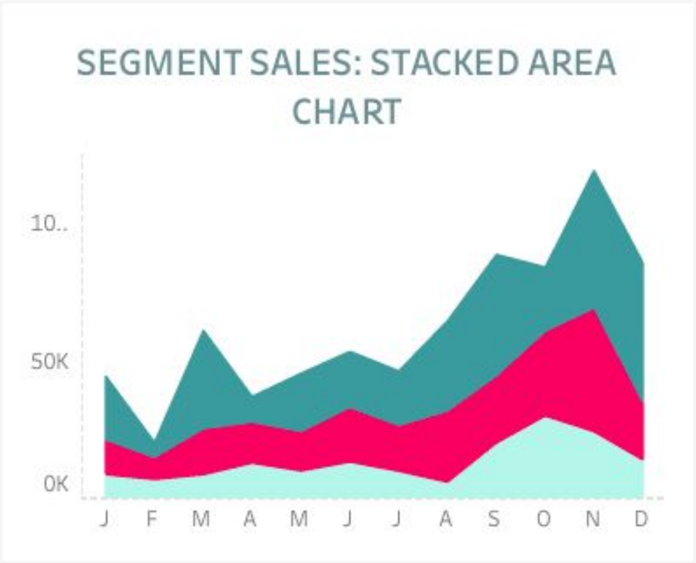
Comparing values: Bar vs Column Chart



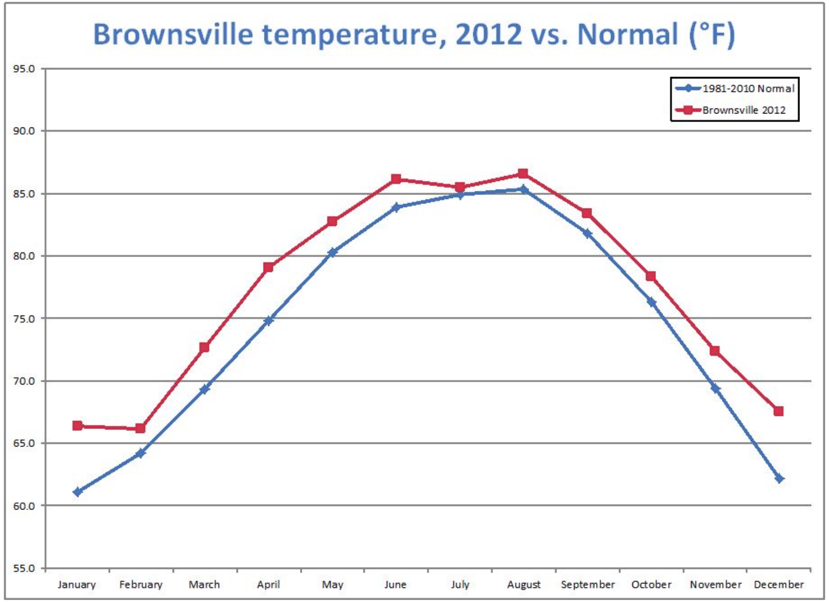


Sales by region

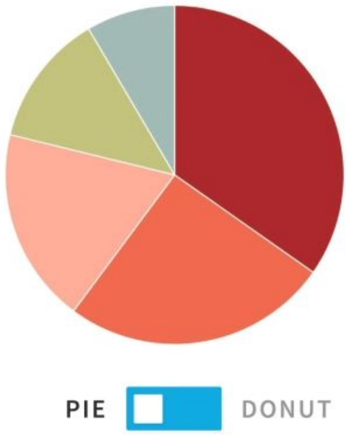
Showing trends: Line vs Area Chart

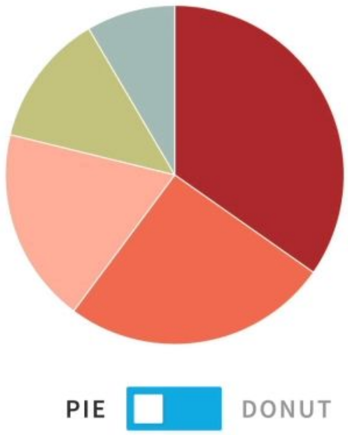


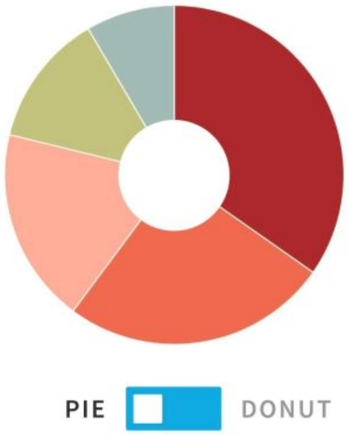
Temperature over time



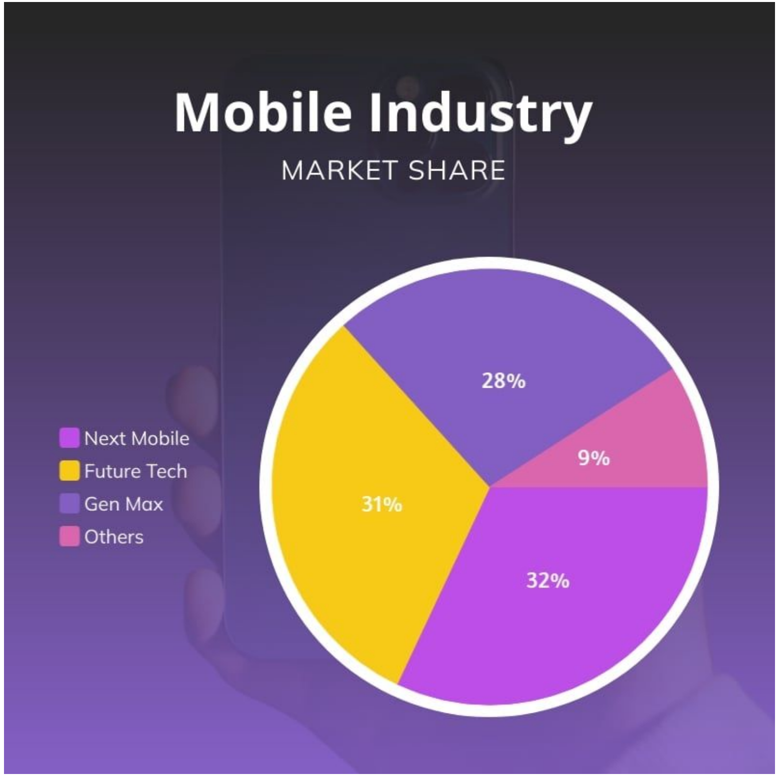
Composition: Pie vs Doughnut chart



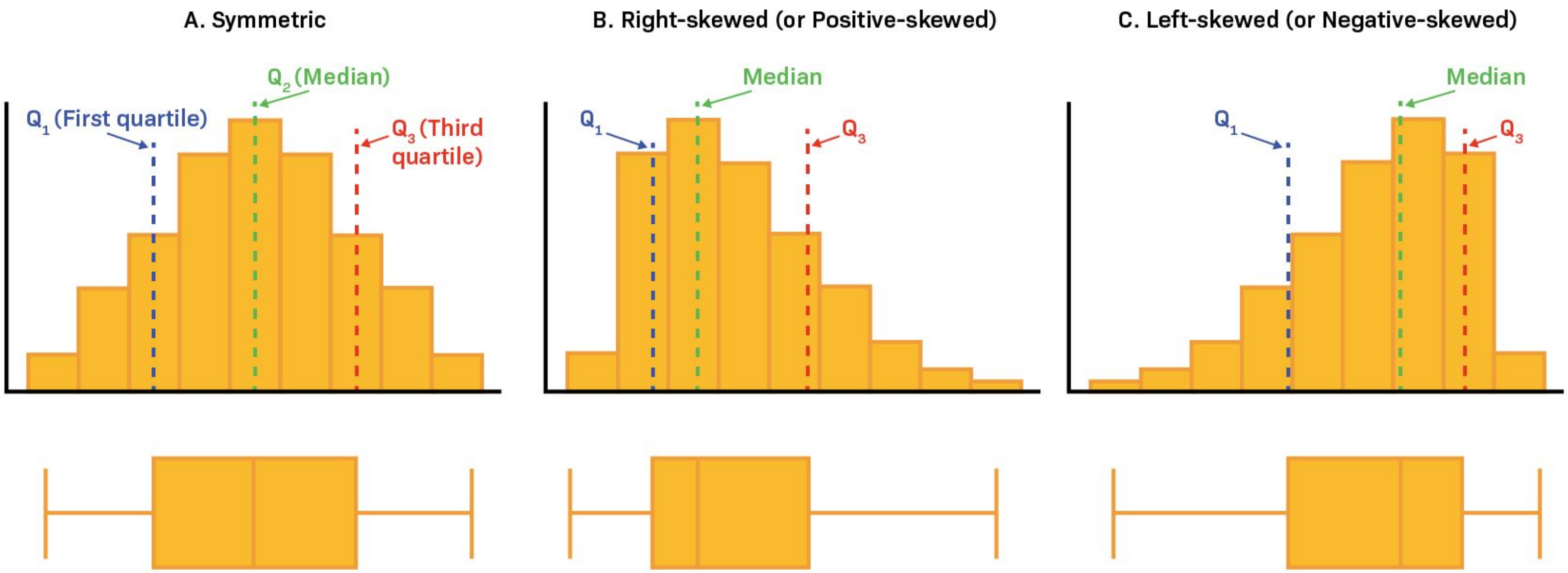




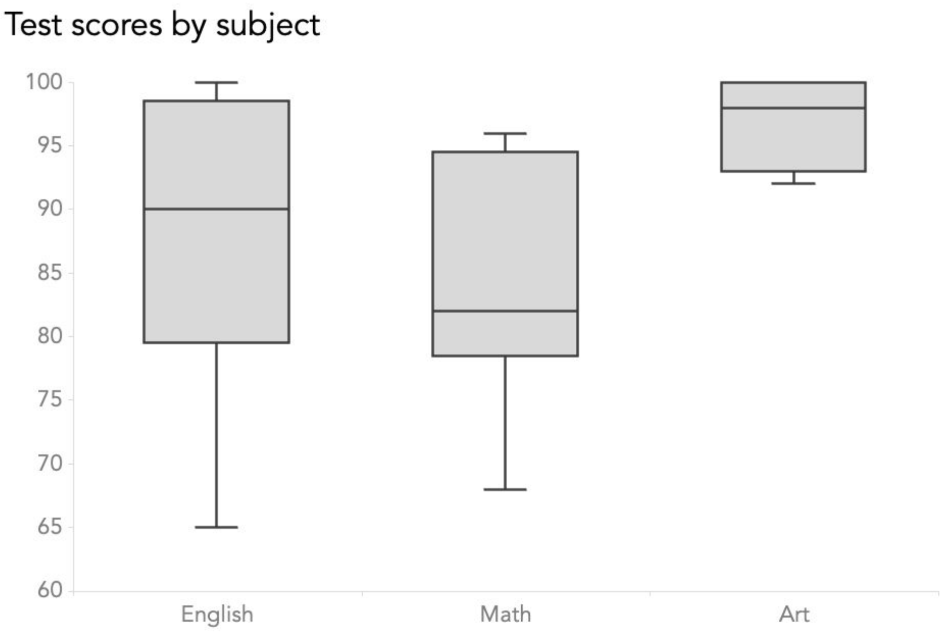
Market share



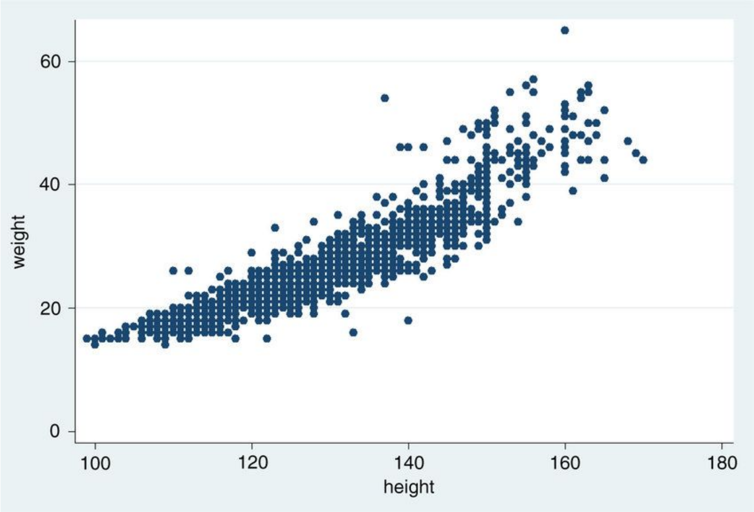
Distribution: Histogram vs Box Plot

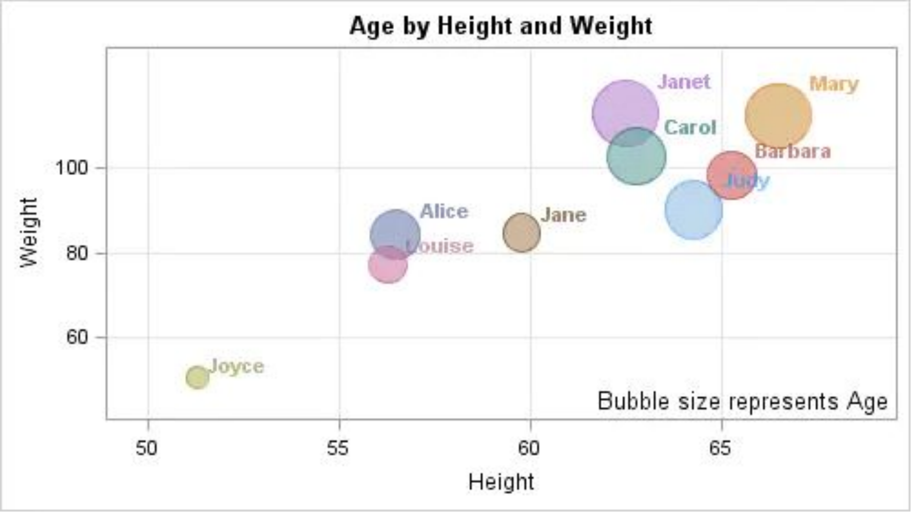


Test scores

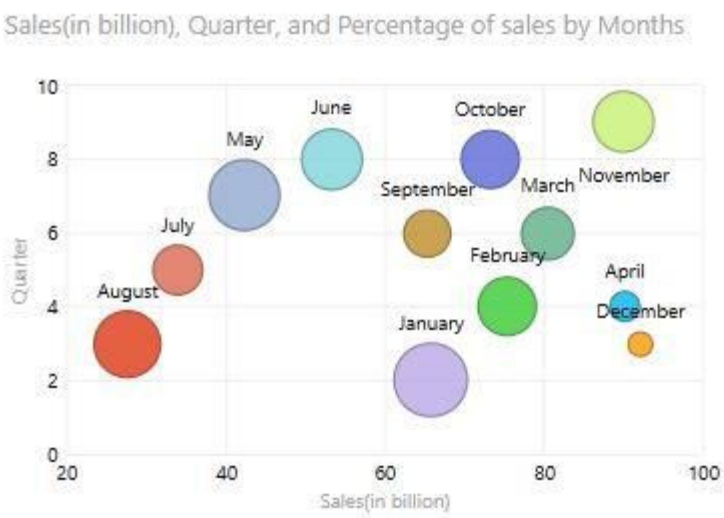

Test scores by subject
100
95
90
85
80
75
70
65
60
English
Math
Art

Relationship: Scatter vs Bubble chart



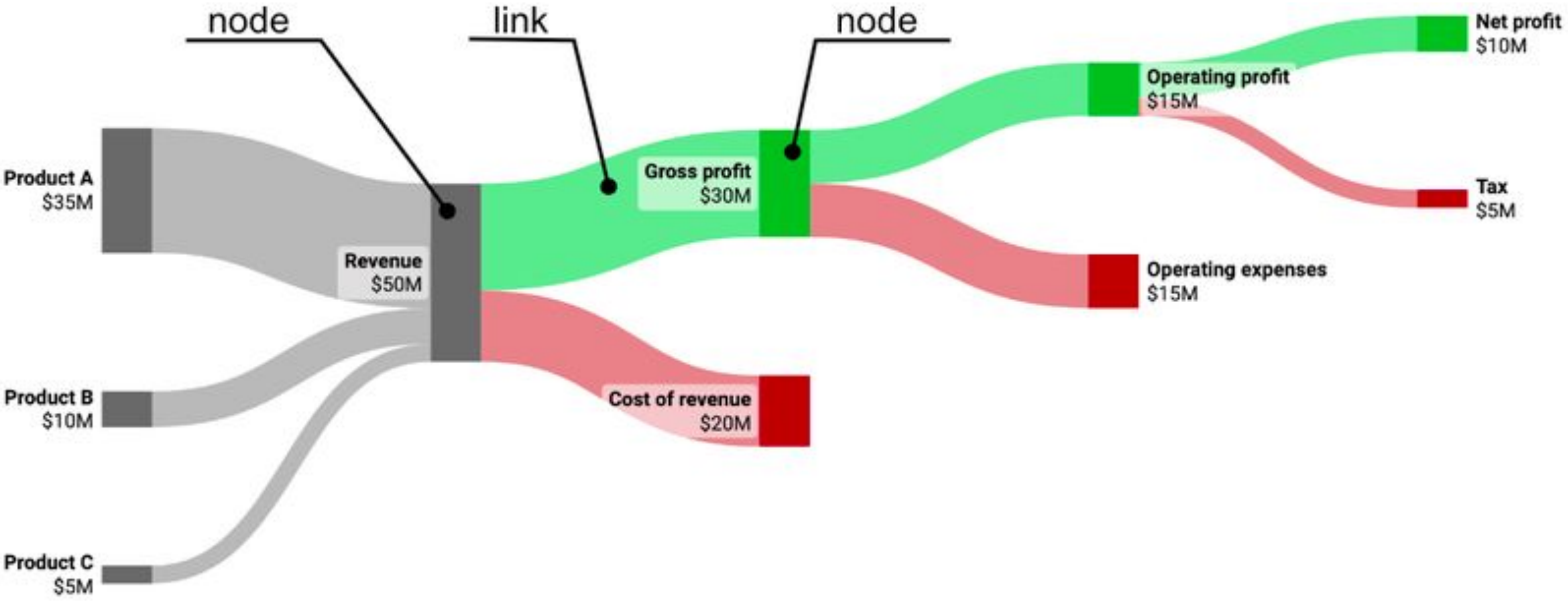


Relationship example

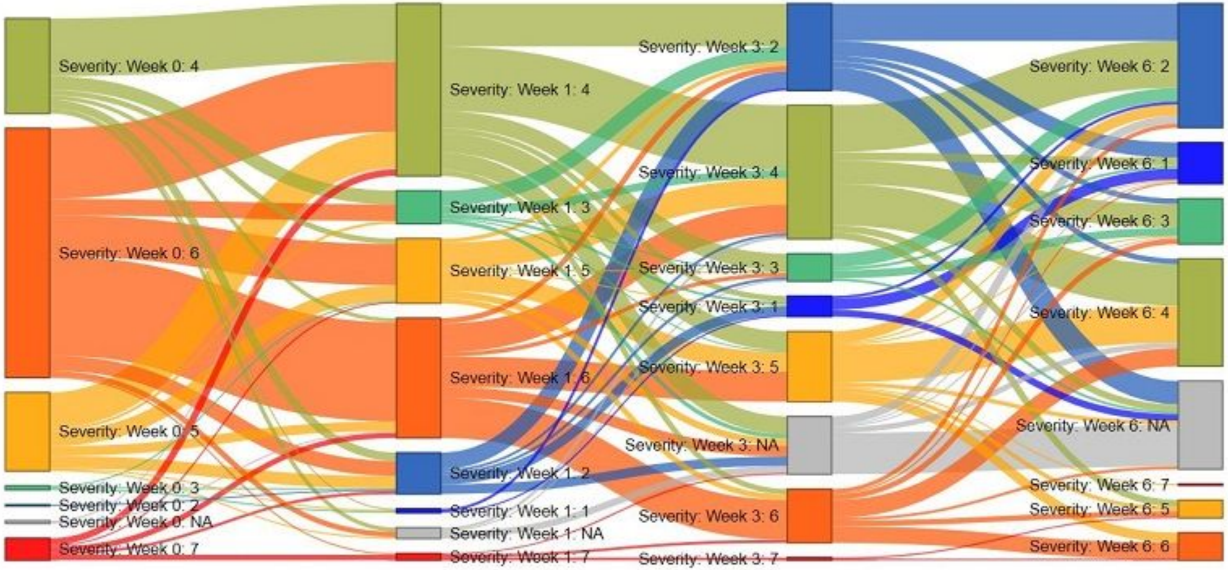


Months	Percentage of sales	Quarter	Sales(in billion)
April	0.10	4	90.20
August	0.65	3	27.50
December	0.05	3	92.00
February	0.50	4	75.30
January	0.80	2	65.80
July	0.35	5	33.90
June	0.55	8	53.30
March	0.40	6	80.50
May	0.75	7	42.30
November	0.55	9	89.90
October	0.50	8	73.30
September	0.30	6	65.20
Total	**5.50**	**65**	**789.20**

Flow: Sankey diagram



Sankey diagram example 2



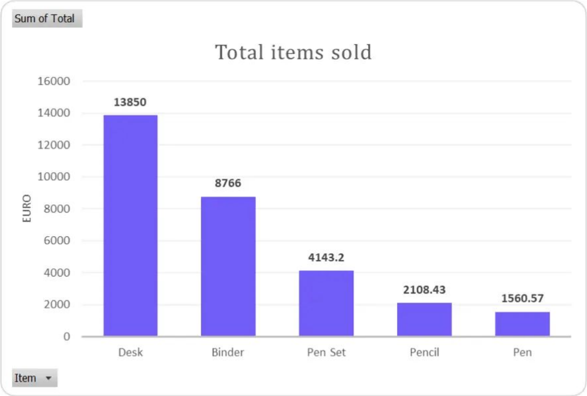
Data visualization process

1. Start with a question
2. Know your audience
3. Choose the right chart type
4. Maximize the data-ink ratio
5. Use color effectively
6. Be honest with scales

Bad vs Good visualizations

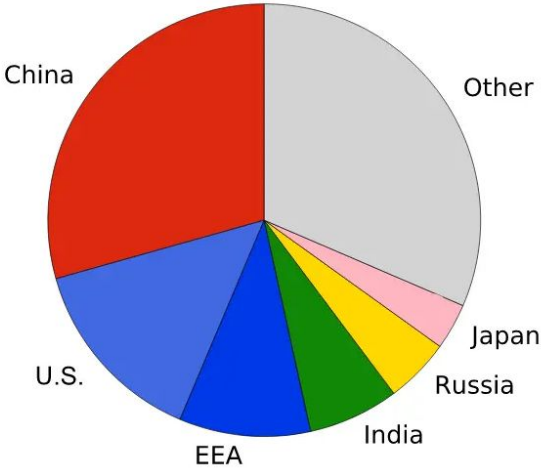
Cluttered chart

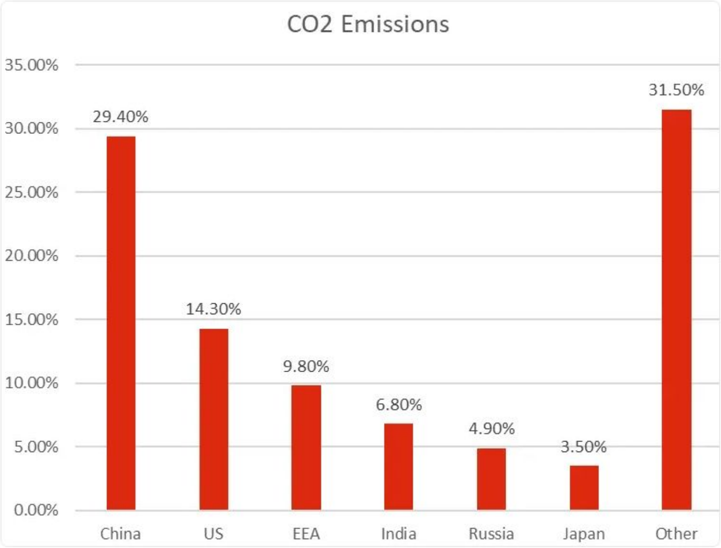
(bad data visualization example)



Clean chart

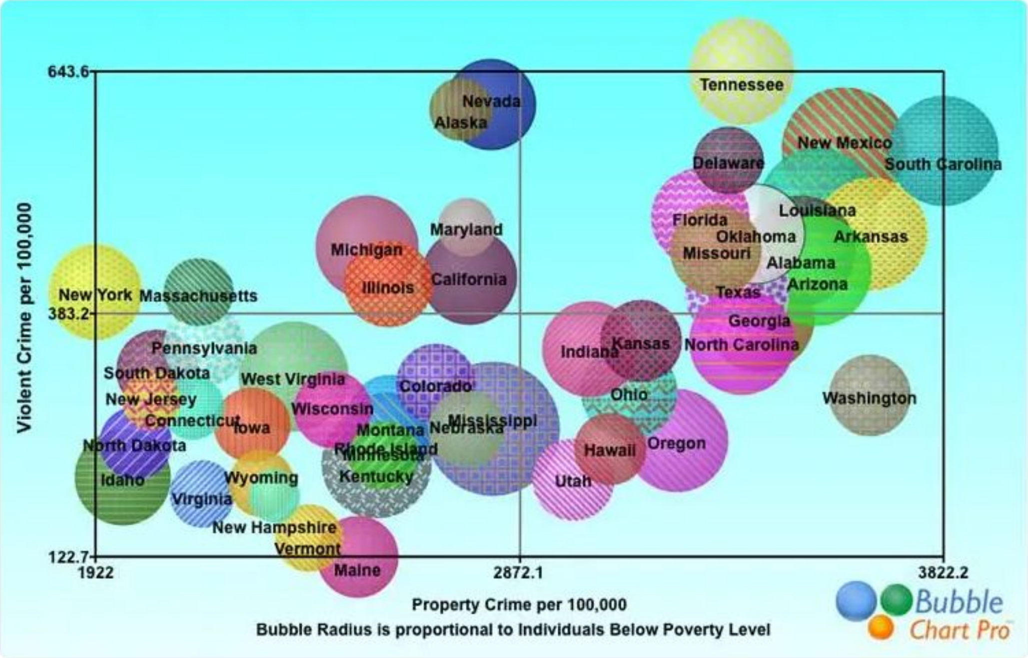
(good data visualization example)


China
Other
Japan
Russia
India
EEA
U.S.


CO2 Emissions
35.00%
30.00%
25.00%
20.00%
15.00%
10.00%
5.00%
0.00%
29.40%
14.30%
9.80%
6.80%
4.90%
3.50%
31.50%
China
US
EEA
India
Russia
Japan
Other


Number of sales for each product
100
95
90
85
80
75
70
Product 1
Product 2


Number of sales for each product
100
90
80
70
60
50
40
30
20
10
0
Product 1
Product 2



643.6
383.2
122.7
1922
2872.1
3822.2
Violent Crime per 100,000
Property Crime per 100,000
Bubble Radius is proportional to Individuals Below Poverty Level
Tennessee
Nevada
Alaska
New Mexico
South Carolina
Delaware
Louisiana
Florida
Oklahoma
Arkansas
Missouri
Alabama
Arizona
Texas
Georgia
North Carolina
Maryland
Michigan
California
Illinois
New York
Massachusetts
Pennsylvania
South Dakota
West Virginia
Colorado
New Jersey
Wisconsin
Connecticut
Iowa
Montana
Nebraska
Mississippi
North Dakota
Rhode Island
Minnesota
Kentucky
Idaho
Wyoming
Virginia
New Hampshire
Vermont
Maine
Indiana
Kansas
Ohio
Washington
Hawaii
Oregon
Utah
Bubble Chart Pro

Some examples of "bad" visualizations: https://viz.wtf/
https://www.jotform.com/blog/bad-data-visualization/

Some examples of "good" visualizations: https://informationisbeautiful.net/

Web visualization libraries

- SVG-based: d3.js, Victory (React), Plotly, ApexCharts
- Canvas-based: Chart.js, ECharts
- Commercial/enterprise: Highcharts (SVG), amCharts

Building a dashboard

A **dashboard** is a visual tool that provides an overview of the most important information, used for monitoring and insight.

It uses **graphs, charts,** and **tables** to present complex data in an understandable format.

Uses:

- ❖ Monitoring performance
- ❖ Spotting trends
- ❖ Decision-making

Key components:

- ❖ KPIs
- ❖ Charts and graphs
- ❖ Filters and controls
- ❖ Layout

Dashboard design process

- ❖ Define objectives
- ❖ Sketch the layout
- ❖ Choose the right visuals
- ❖ Implement and iterate

Business Intelligence Platforms

- Tableau
- Power BI
- Looker Studio

Visualization in MVC app

- ❖ **User Action:** The Controller receives a request (e.g., load page, apply filter).
- ❖ **Controller -> Model:** The Controller asks the Model for data.
- ❖ **Model -> Controller:** The Model fetches/processes data (e.g., from a GET endpoint) and returns it.
- ❖ **Controller -> View:** The Controller passes the data to the View.
- ❖ **View Renders:** The View initializes the chart library (Chart.js, D3) and renders the visualization with the new data.